

Shrinkage Methods

Ridge, LASSO, and the Bias–Variance Tradeoff

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Roadmap

1. Motivation: When p Is Large
2. Best Subset Selection
3. The Bias–Variance Tradeoff
4. Ridge Regression
5. The LASSO
6. Worked Example: Used-Car Prices
7. Summary

1. Motivation: When p Is Large

1.1 Recall: R^2 Always Increases with p

From Lecture 9 §6, the multiple regression R^2 is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

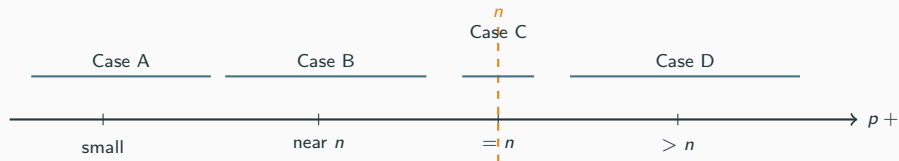
Adding any predictor (even a useless one) can only *lower* RSS, so R^2 never decreases.

The tempting conclusion: “more predictors $\Rightarrow$ bigger $R^2 \Rightarrow$ better model.” Should we just throw every variable we have into $\mathbf{X}$?

Today's lecture is the reply. We will see that for large p , OLS itself breaks down — and that the fix is not to enumerate models but to *penalize* the coefficients.

1.2 Four Regimes of n vs $p + 1$

Recall: the design matrix $\mathbf{X}$ is $n \times (p + 1)$ (intercept column plus p predictors). The behaviour of OLS depends on how n compares with the number of columns $p + 1$.



- **Case A:** $n \gg p + 1$ — the classical world.
- **Case B:** $p + 1 < n$ but close — variance trouble.
- **Case C:** $p + 1 = n$ — exact interpolation.
- **Case D:** $p + 1 > n$ — the **high-dimensional** case.

1.3 Case A — $n \gg p + 1$

Plenty of rows, few columns. The design matrix is tall and thin:



- $\mathbf{X}^\top \mathbf{X}$ is invertible (with no perfect multicollinearity).
- OLS exists, is unique, and has small variance:

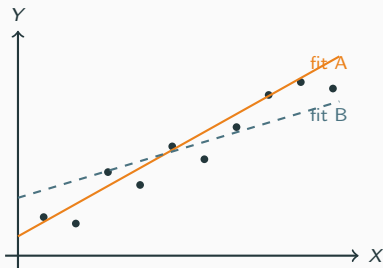
$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

- This is the world Lectures 6–9 lived in: n in the hundreds or thousands, p no more than a handful.

1.4 Case B — $p + 1 < n$ but Close

Now $p + 1$ is large relative to n . OLS still exists, but $(\mathbf{X}^\top \mathbf{X})^{-1}$ has large entries, and the estimated coefficients become unstable.

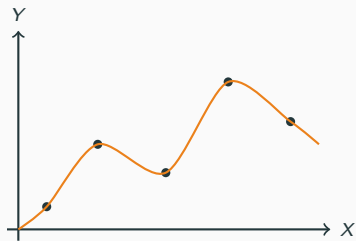
Visual. Two random subsamples of the same population produce wildly different fitted lines:



Each $\hat{\beta}_j$ has a large standard error (L9 §3); predictions are noisy. OLS is unbiased, but variance dominates.

1.5 Case C — $p + 1 = n$

Number of parameters equals number of equations. OLS fits the data *exactly*: every residual is zero.



Symptoms:

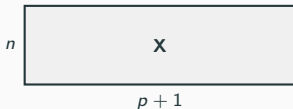
- $RSS = 0 \Rightarrow R^2 = 1$ trivially.
- No degrees of freedom left to estimate σ^2 .

1.5 Case C — $p + 1 = n$ 2

- Inference machinery (CI, t-test, F-test) collapses.
- The fit memorises the training data; it will generalise poorly to new $\mathbf{x}_{\text{new}}$.

1.6 Case D — $p + 1 > n$ (High-Dimensional)

More columns than rows. The design matrix is short and wide:



Key Result — OLS breaks down

When $p + 1 > n$:

- $\mathbf{X}^\top \mathbf{X}$ is **singular** (cannot be inverted).
- The OLS normal equations have **infinitely many** solutions β that drive RSS to zero.
- There is no way to pick one of them from the data alone.

We call this the **high-dimensional** case. It is the setting where modern data lives.

1.7 Where High-Dimensional Data Arises

Example 1 — DNA / gene-expression studies

A medical study measures expression levels of $p \approx 20,000$ genes in $n \approx 100$ patients to predict a disease outcome Y . Then $p \gg n$ — hundreds of times more columns than rows. OLS cannot run at all; we need a method that identifies which genes carry predictive signal.

Example 2 — Image recognition

A 128×128 grayscale image gives $p = 16,384$ pixel predictors. A labelled training set of a few hundred images is common. Again $p \gg n$. Plus most pixels are uninformative background — we want a method that automatically ignores them.

These are not corner cases; they are the routine setting for genomics, computer vision, text, and finance.

1.8 What This Lecture Delivers

We need two ideas:

1. **Variable selection.** Pick a subset of predictors and drop the rest.
2. **Shrinkage / regularization.** Keep all predictors but pull their slope coefficients toward zero.

Roadmap for the rest of the lecture:

- §2: best subset selection — a clean idea that does not scale.
- §3: bias-variance tradeoff — why a biased estimator can beat OLS.
- §4: ridge regression — ℓ_2 penalty, closed form.
- §5: LASSO — ℓ_1 penalty, automatic variable selection.
- §6: worked example with simulated software output.

2. Best Subset Selection

2.1 Best Subset Selection — The Idea

The most direct response to “which predictors matter?” is to try every possible subset.

Definition — best subset selection

Given predictors $X_1, \dots, X_p$, for each $k = 0, 1, \dots, p$:

1. Fit every model that uses exactly k of the predictors — there are $\binom{p}{k}$ such models.
2. Among them, keep the one with the smallest RSS. Call it $\mathcal{M}_k$.

Then pick the best k across $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using a criterion like *adjusted R^2* , AIC, or validation MSE.

Notation reminder. $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ counts the number of ways to pick k items from p .

2.2 Worked Example — $p = 3$ Predictors

Setup. Three candidate predictors: X_1, X_2, X_3 . With $p = 3$, total models $= 2^3 = 8$ (each X_j is in or out).

Fabricated training RSS for each subset:

Predictors in model	k	RSS (training)
none (intercept only)	0	850
$\{X_1\}$	1	620
$\{X_2\}$	1	720
$\{X_3\}$	1	810
$\{X_1, X_2\}$	2	410
$\{X_1, X_3\}$	2	580
$\{X_2, X_3\}$	2	690
$\{X_1, X_2, X_3\}$	3	405

2.2 Worked Example — $p = 3$ Predictors 2

Adding X_3 to $\{X_1, X_2\}$ barely lowers RSS (from 410 to 405). Adjusted R^2 prefers $\mathcal{M}_2 = \{X_1, X_2\}$ — it pays a complexity tax for the extra predictor that the small RSS gain does not justify. Best subset's verdict: $\{X_1, X_2\}$.

2.3 The 2^p Blow-Up

Each predictor is either in or out, so the total number of subsets to fit is

$$\sum_{k=0}^p \binom{p}{k} = 2^p.$$

p	2^p	comment
10	1,024	easy
20	$\sim 10^6$	slow but possible
30	$\sim 10^9$	hours on a laptop
50	$\sim 10^{15}$	beyond a single machine
100	$\sim 10^{30}$	hopeless
20,000	$\sim 10^{6000}$	more than atoms in the universe

Computational verdict. Best subset is infeasible once p is moderate. The genomic example ($p \approx 20,000$) is comically out of reach.

2.4 Even Free Compute Does Not Save Us

Suppose, hypothetically, that we could fit all 2^p models. There is still a statistical problem.

- Each subset comparison is an implicit hypothesis test.
- With 2^p comparisons, the chance that one model looks good *by sheer luck* is essentially 1.
- The “best” subset overfits the selection step itself — it picks up sample-specific noise that will not reappear on new data.

So best subset has two strikes: it is computationally infeasible, and even if it were not, it would be statistically unreliable.

2.5 What We Need Instead

We need a procedure that picks variables *as part of fitting one model*, not by enumerating 2^p separate models.

The plan

Modify the OLS objective so the fit *itself* prefers smaller coefficients — and, ideally, drives some of them exactly to zero. One optimisation, one model.

Before introducing the methods, we need one more tool: a precise way to compare estimators that have different bias and variance. That is the next section.

3. The Bias–Variance Tradeoff

3.1 Setup — A General Target

Consider *any* fixed (non-random) quantity θ we want to learn, and *any* estimator $\hat{\theta}$ we build from training data.

Notation reminder. θ is the **target**; it can be many different objects, e.g.:

- a coefficient: $\theta = \beta_j$;
- a regression-function value at a new point: $\theta = f(\mathbf{x}_0)$.

$\hat{\theta}$ is our estimator of θ — a function of the training data, hence *random*.

We measure accuracy by the **mean squared error**

$$\text{MSE}(\hat{\theta}) \equiv \mathbb{E}[(\hat{\theta} - \theta)^2].$$

Goal: decompose this expectation into interpretable pieces.

3.2 The Bias–Variance Identity

Let $\mu = \mathbb{E}[\hat{\theta}]$ — the average of $\hat{\theta}$ across hypothetical training samples.
Add and subtract μ :

$$\hat{\theta} - \theta = \underbrace{(\hat{\theta} - \mu)}_{\text{random, mean 0}} + \underbrace{(\mu - \theta)}_{\text{constant}}.$$

Square and take expectation:

$$\begin{aligned}\mathbb{E}[(\hat{\theta} - \theta)^2] &= \mathbb{E}[(\hat{\theta} - \mu)^2] + 2(\mu - \theta) \underbrace{\mathbb{E}[\hat{\theta} - \mu]}_{=0} + (\mu - \theta)^2 \\ &= \text{Var}(\hat{\theta}) + (\mu - \theta)^2.\end{aligned}$$

3.2 The Bias–Variance Identity 2

Key Result — bias–variance identity

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{Bias}^2(\hat{\theta})} + \underbrace{\text{Var}(\hat{\theta})}_{\text{Variance}}.$$

Two terms only. The first measures systematic error (how far the average estimator is from the truth); the second measures wobble (how much $\hat{\theta}$ varies across training samples).

3.3 Two Concrete Instantiations of θ

The identity in §3.2 specialises in two useful ways:

Target θ	Estimator $\hat{\theta}$	MSE =
β_j (a coefficient)	$\hat{\beta}_j$	$\text{Bias}(\hat{\beta}_j)^2 + \text{Var}(\hat{\beta}_j)$
$f(\mathbf{x}_0)$ (regression value)	$\hat{f}(\mathbf{x}_0)$	$\text{Bias}(\hat{f}(\mathbf{x}_0))^2 + \text{Var}(\hat{f}(\mathbf{x}_0))$

Whichever target we care about, the two pieces of MSE are **always** the same: Bias² and Variance. The rest of the lecture is about trading them off.

3.4 Reading the Decomposition

Both pieces — Bias² and Variance — are under our control through the choice of estimator.

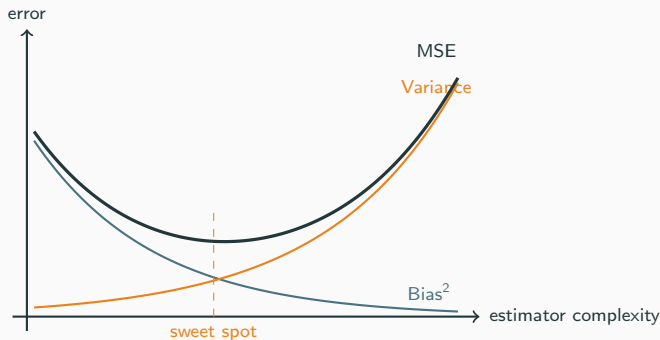
- Simple estimator (e.g., always predict $\bar{y}$): high bias, low variance.
- Complex estimator (e.g., OLS when p is large relative to n): low bias, high variance.
- The best estimator trades some of one for the other.

The big idea. A *biased* estimator can have a smaller MSE than an unbiased one, if the bias it adds is more than compensated by the variance it removes. This breaks the reflex that an unbiased estimator must be the best one.

Ridge and LASSO are exactly such estimators: they introduce a small bias and buy back a lot of variance.

3.5 The Tradeoff Picture

Plot MSE and its two components as a function of estimator complexity:



MSE has a U-shape with a clear minimum — the optimal complexity. OLS when p is large relative to n sits far to the right of the minimum (high variance). Ridge and LASSO let us walk *back left* by tuning a single parameter λ .

3.6 OLS in This Framework

Under the L9 assumptions, OLS is *unbiased*:

$$\mathbb{E}[\hat{\beta}_{\text{OLS}} \mid \mathbf{X}] = \beta \implies \text{Bias} = 0.$$

So all of OLS's MSE for β is variance:

$$\text{Var}(\hat{\beta}_{\text{OLS}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

where $\sigma^2 = \text{Var}(\epsilon)$ is the noise variance of the underlying regression model (L9).

When $\mathbf{X}^\top \mathbf{X}$ is near-singular — Case B from §1.4 — this variance blows up. The diagonal of $(\mathbf{X}^\top \mathbf{X})^{-1}$ contains huge numbers.

Take away

For large p , OLS sits at the **far end** of the bias-variance tradeoff: zero bias, big variance. Adding a small bias is the **cheapest way** to lower variance and improve MSE.

4. Ridge Regression

4.1 The Idea

OLS minimises RSS with no restriction on β . If $\mathbf{X}^\top \mathbf{X}$ is near-singular, the optimal β can have enormous entries that exactly cancel on training data but generalise poorly.

Ridge regression penalises the size of the slope coefficients using the squared ℓ_2 idea from L8 §5:

$$\sum_{j=1}^p \beta_j^2.$$

Large slopes now cost us; the optimiser is pushed toward smaller, more stable values. Ridge and LASSO shrink slopes $\beta_1, \dots, \beta_p$, not the intercept β_0 .

4.2 Definition

Definition — ridge regression

Given a hyperparameter $\lambda \geq 0$, the **ridge** estimator solves

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta \in \mathbb{R}^{p+1}} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}.$$

Notation reminder. The objective is a function $L_\lambda : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$,
 $L_\lambda(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p \beta_j^2$. We minimise it over β .

4.2 Definition 2

λ is a **hyperparameter**

Unlike β , the number $\lambda \geq 0$ is *not* estimated by minimising the objective — it sits *inside* the objective and controls how aggressively we shrink. A free parameter we have to fix from outside. **For now, treat λ as given.**

Whatever value we plug in, the ridge solution is well-defined. At the end of §4 (§4.8) we come back and show how to pick λ from the data itself, using a procedure called **cross-validation**.

Two extremes:

- $\lambda = 0$: penalty disappears, $\hat{\beta}_{\text{ridge}} = \hat{\beta}_{\text{OLS}}$.
- $\lambda \rightarrow \infty$: the fitted slopes $(\hat{\beta}_1, \dots, \hat{\beta}_p)$ approach **0** and the model approaches an intercept-only fit.

In between, ridge shrinks the OLS slopes toward zero in a smooth, controlled way while leaving the intercept unpenalized.

4.3 Matrix Derivation on Centered Data

For this derivation only, center $\mathbf{y}$ and every predictor. The intercept is handled separately and recovered afterward as $\hat{\beta}_0 = \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{x}_j$. Below, $\mathbf{X}$ is the $n \times p$ centered slope design and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ contains slopes only.

Write the centered ridge objective in matrix form and minimise. Four lines.

Step 1 — expand. Using $\|\mathbf{v}\|_2^2 = \mathbf{v}^\top \mathbf{v}$,

$$L_\lambda(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta}.$$

Step 2 — distribute.

$$L_\lambda(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta}.$$

4.3 Matrix Derivation on Centered Data 2

Step 3 — take the gradient (L9 Appendix A rules: $\nabla_{\beta}(\mathbf{a}^{\top}\beta) = \mathbf{a}$ and for symmetric $\mathbf{A}$, $\nabla_{\beta}(\beta^{\top}\mathbf{A}\beta) = 2\mathbf{A}\beta$):

$$\nabla_{\beta}L_{\lambda} = -2\mathbf{X}^{\top}\mathbf{y} + 2\mathbf{X}^{\top}\mathbf{X}\beta + 2\lambda\beta.$$

Step 4 — set to zero and solve.

$$(\mathbf{X}^{\top}\mathbf{X} + \lambda\mathbf{I}_p)\beta = \mathbf{X}^{\top}\mathbf{y}.$$

4.4 The Ridge Estimator

Continue the centered-data convention from §4.3: β contains the p slopes only, and the intercept is recovered separately.

Key Result — ridge estimator

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^{\top} \mathbf{y}$$

Side-by-side with OLS from L9:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{X}^{\top} \mathbf{y}, \quad \hat{\beta}_{\text{ridge}} = (\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^{\top} \mathbf{y}.$$

The only change is the addition of $\lambda \mathbf{I}_p$ inside the inverse. As $\lambda \rightarrow 0$ we recover OLS; as $\lambda \rightarrow \infty$ the inverse approaches $\frac{1}{\lambda} \mathbf{I}_p$ and the slope vector shrinks to $\mathbf{0}$. The intercept remains separate and unpenalized.

4.5 Why Ridge Always Exists

Under the centered slope convention from §4.3, when $p + 1 > n$ (equivalently $p \geq n$), $\mathbf{X}^\top \mathbf{X}$ is singular and OLS fails.

Key Result — ridge fixes the singularity

For any $\lambda > 0$, the matrix

$$\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p$$

is invertible — regardless of n , p , or multicollinearity.

Intuition: adding $\lambda \mathbf{I}_p$ bumps every diagonal entry up by λ , which lifts the smallest eigenvalues away from zero. The intercept is handled separately, and the computer reports the slope solution without difficulty.

Consequence. Ridge applies to all four regimes of §1, including the high-dimensional case ($p + 1 > n$). OLS only applies to Cases A–B.

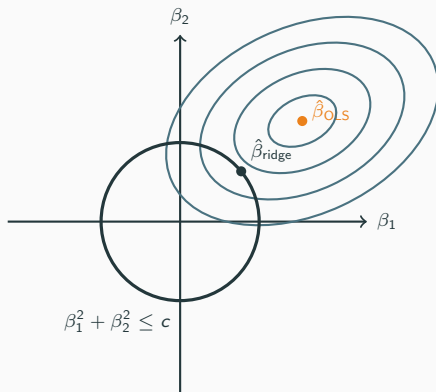
4.6 Constraint View — The Geometry

An equivalent way to write ridge: for each λ there is a corresponding budget c such that

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq c.$$

Picture in the (β_1, β_2) plane:

4.6 Constraint View — The Geometry 2



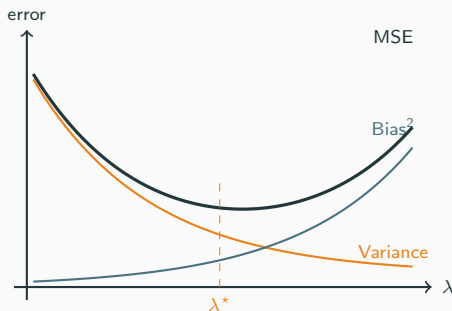
4.6 Constraint View — The Geometry 3

Ellipses are level sets of $\text{RSS}(\beta)$; the disc is the slope-constraint region; the intercept is not shown and is unpenalized. The ridge solution is where the smallest ellipse that touches the disc meets it — generically *not* on an axis.

4.7 Ridge in the Bias–Variance Picture

As we crank λ up from 0 to ∞ :

- bias *grows* (the slope estimator is pulled away from the true slope vector),
- variance *shrinks* (the answer becomes more stable),
- MSE follows a U-shape with a clear minimum.



4.7 Ridge in the Bias–Variance Picture 2

$\lambda = 0$ is OLS (left edge); large λ pushes the slopes toward zero and the fit toward the intercept-only model. The minimum λ^* is the goal.

4.8 Choosing λ

Time to deliver on the promise made back in §4.2: how to pick the hyperparameter λ from the data, rather than by hand.

Forward pointer to Lecture 13

Cross-validation estimates the test MSE at a grid of λ values and picks the λ with the smallest CV error. Software handles the entire procedure.

For now, treat λ as a tuning parameter whose value comes from a procedure we will study in L13. The qualitative picture is what matters here: bigger $\lambda \Rightarrow$ more shrinkage $\Rightarrow$ more bias and less variance.

Reminder. Ridge and LASSO shrink the *slope* coefficients $\beta_1, \dots, \beta_p$ but leave the intercept β_0 alone. Most software also standardises the predictors first (mean 0, variance 1) so that the penalty is scale-fair. “Computer does it.”

5. The LASSO

5.1 From ℓ_2 to ℓ_1

Ridge shrinks every slope toward zero, but rarely to zero. All p predictors stay in the model.

For interpretation we often prefer a **sparse** model:

- Only a handful of predictors should remain.
- The rest should be dropped automatically.

Idea: swap the ℓ_2 penalty for the ℓ_1 penalty from L8 §5:

$$\sum_{j=1}^p |\beta_j|.$$

Same kind of objective, different penalty.

5.2 Definition — the LASSO

Definition — LASSO

Given a hyperparameter $\lambda \geq 0$, the **LASSO** (Least Absolute Shrinkage and Selection Operator) solves

$$\hat{\beta}_{\text{LASSO}} = \arg \min_{\beta \in \mathbb{R}^{p+1}} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}.$$

Same hyperparameter convention as ridge (§4.2): treat λ as given for now; we will pick it by cross-validation at the end of the section, just as we did for ridge.

Equivalent constraint form:

$$\hat{\beta}_{\text{LASSO}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq c.$$

5.2 Definition — the LASSO 2

Same shape as ridge — squared slope penalties become absolute slope penalties. The intercept remains outside the constraint.

5.3 Historical Note

Where the LASSO came from

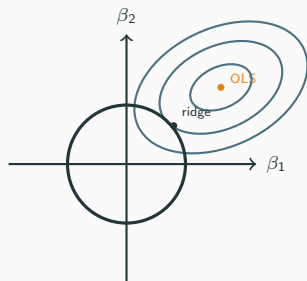
The LASSO was introduced to the statistics community by Robert Tibshirani in 1996 — one of the authors of our textbook (ISLR).

It is widely considered one of the most influential ideas in modern statistics. A single change of penalty (from ℓ_2 to ℓ_1) turns OLS into a method that performs variable selection *automatically*, in one fit, even when $p > n$ — a regime where OLS itself does not exist.

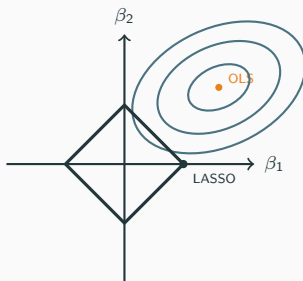
The next few slides explain *why* an ℓ_1 penalty has this effect, while ℓ_2 does not.

5.4 Why LASSO Sets Slopes to Zero

Compare the two constraint regions in the (β_1, β_2) plane. RSS contours (ellipses) are the same for both methods.



$$\text{Ridge: } \beta_1^2 + \beta_2^2 \leq c$$



$$\text{LASSO: } |\beta_1| + |\beta_2| \leq c$$

The ℓ_1 diamond has **corners on the axes**. RSS ellipses generically touch the diamond *at a corner*, which is exactly the event that some $\hat{\beta}_j$ equals zero. The ℓ_2 disc is smooth and has no such corners.

5.5 Variable Selection as a Side Effect

Key Result — automatic selection

For λ large enough, some of the $\hat{\beta}_j^{\text{LASSO}}$ are **exactly zero**. The predictors with $\hat{\beta}_j = 0$ drop out of the model.

Compare with the procedures we have seen:

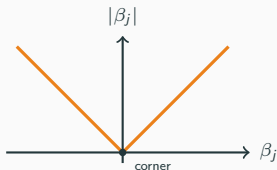
- **Best subset (§2)**: 2^p fits — infeasible.
- **Ridge (§4)**: all p predictors retained — no selection.
- **LASSO**: *one* fit, and some predictors are dropped *automatically* as a byproduct of the optimisation.

This is the answer L9 §8 promised: variable selection performed as part of the fit, not before it.

5.6 No Closed Form

Ridge had a closed-form solution because both $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$ and $\lambda \sum_{j=1}^p \beta_j^2$ are differentiable in $\boldsymbol{\beta}$. Set the gradient to zero, solve.

The LASSO penalty $|\beta_j|$ is **not differentiable at $\beta_j = 0$** — it has a sharp corner there:



Calculus cannot set “the derivative” to zero at that corner. Specialised optimisation algorithms (coordinate descent, proximal-gradient) are used instead.

Bottom line. “Computer does it.” The software returns $\hat{\boldsymbol{\beta}}_{\text{LASSO}}$ as a numeric vector — with some slope entries equal to zero.

5.7 OLS vs Ridge vs LASSO

Three estimators of the same coefficients β , compared head-to-head:

	OLS	Ridge	LASSO
Penalty	none	$\lambda \sum_{j=1}^p \beta_j^2$	$\lambda \sum_{j=1}^p \beta_j $
Closed form?	yes	yes	no
Bias	0	small, grows with λ	small, grows with λ
Variance	large for large p	reduced	reduced
Sets a slope to 0?	no	no	yes
Works if $p + 1 > n$?	no	yes	yes

When to use which.

- OLS: p small, all predictors plausibly relevant.

5.7 OLS vs Ridge vs LASSO 2

- Ridge: p moderate-to-large, want stable prediction, do not care about dropping variables.
- LASSO: p large, suspect that only a few predictors carry signal, want a sparse and interpretable model.

6. Worked Example: Used-Car Prices

6.1 Setup

A dealership wants to predict the sale price of a used car from its features. We have $n = 200$ training cars with $p = 8$ predictors:

Variable	Type	Description
X_1 mileage	continuous	odometer reading (1000 mi)
X_2 age	continuous	years since manufacture
X_3 mpg_city	continuous	city fuel economy
X_4 mpg_hwy	continuous	highway fuel economy
X_5 horsepower	continuous	engine horsepower
X_6 n_owners	integer	number of prior owners
X_7 accident	0/1	history of major accident?
X_8 luxury	0/1	luxury brand?

Response Y = sale price (USD). We split the data 160 training / 40 validation and compare best subset, OLS, ridge, and LASSO on validation MSE.

6.2 Best Subset Selection — Software Output

For each model size $k = 1, \dots, 8$, the algorithm fits all $\binom{8}{k}$ subsets and returns the one with the smallest RSS. (We skip the trivial $k = 0$ intercept-only baseline, whose $\text{RSS} = \text{TSS}$ by construction.) The training split has $n = 160$ and $\text{TSS} = 4.70 \times 10^9$. Simulated output:

k	best subset of predictors	RSS	adj. R^2
1	age	1.820×10^9	0.610
2	age, mileage	1.100×10^9	0.763
3	age, mileage, horsepower	0.810×10^9	0.824
4	age, mileage, horsepower, luxury	0.620×10^9	0.865
5	+ accident	0.618×10^9	0.864
6	+ mpg_hwy	0.617×10^9	0.864
7	+ n_owners	0.616×10^9	0.863
8	all eight	0.615×10^9	0.862

6.2 Best Subset Selection — Software Output 2

Adjusted R^2 peaks at $k = 4$: {age, mileage, horsepower, luxury}. Adding more predictors barely lowers RSS and pays a complexity tax. Validation MSE on this best-4 model: 3.9×10^6 .

6.3 OLS, Ridge, and LASSO — Coefficients

Fit all three methods on the full 8-predictor design matrix. Software reports (rounded; intercept omitted):

Coefficient (USD per unit)	OLS	Ridge ($\lambda = 100$)	LASSO ($\lambda = 100$)
mileage	-280	-240	-235
age	-1,450	-1,210	-1,380
mpg_city	-95	-30	0
mpg_hwy	+110	+45	0
horsepower	+85	+72	+68
n_owners	-450	-210	0
accident	-1,900	-1,550	-1,620
luxury	+5,800	+4,900	+5,100

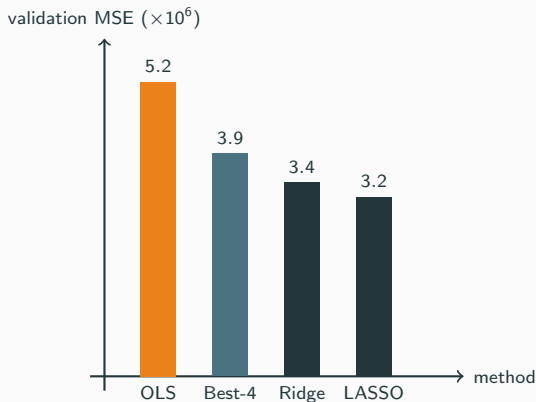
Observations:

6.3 OLS, Ridge, and LASSO — Coefficients 2

- OLS keeps every variable, including the noisy mpg pair.
- Ridge shrinks every slope coefficient toward zero — none exactly zero.
- LASSO sets three coefficients (mpg_city, mpg_hwy, n_owners) to **exactly zero**. Five predictors survive, matching the best-subset selection plus accident.

6.4 Validation MSE Comparison

Same 40 held-out cars; predict price, compute squared errors:



6.4 Validation MSE Comparison 2

Reading. OLS overfits the eight predictors; its validation error is the worst. Best-subset with $k = 4$ already helps. Ridge does a little better by retaining (and shrinking) all eight. LASSO does best here: it retains the same five predictors with shrinkage, dropping three noise variables.

All three regularised methods beat unrestricted OLS — exactly the bias-variance prediction from §3.

6.5 Take-Aways from the Example

- **Best subset** is fine here ($p = 8 \Rightarrow 2^8 = 256$ models). It would be hopeless if p were 1000.
- **Ridge** keeps every variable in the model; useful if we believe many predictors carry some signal.
- **LASSO** reaches a sparse model in one fit; useful when most predictors are uninformative.
- All regularised methods beat OLS on test error — the bias they add is more than paid back by the variance they remove.

Software-wise: every modern statistical package fits ridge and LASSO out of the box; the user supplies λ or asks cross-validation to pick one. We will see how in L13.

7. Summary

7.1 Take Away — Foundations

Three big ideas

- **The R^2 trap.** More predictors always raises R^2 , but OLS breaks down when p is large relative to n , and fails outright when $p + 1 > n$. The high-dimensional regime ($p + 1 > n$) is the routine modern setting.
- **Bias-variance tradeoff.** For any target θ and estimator $\hat{\theta}$, $\text{MSE} = \text{Bias}^2 + \text{Variance}$. A *biased* estimator can have a smaller MSE than an unbiased one if it cuts variance enough.
- **Regularization beats enumeration.** Best subset needs 2^p fits; ridge and LASSO need *one*. LASSO even performs variable selection automatically as a side effect of the ℓ_1 penalty.

7.2 Take Away — Key Formulas

The L10 toolbox

- **OLS:** $\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.
- **Ridge:** $\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \{\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p \beta_j^2\}$; the centered-data slope formula is in §4.4.
- **LASSO:** $\hat{\beta}_{\text{LASSO}} = \arg \min_{\beta} \{\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j|\}$; no closed form, software returns it numerically with some slope entries equal to zero. Both methods leave β_0 unpenalized.
- **Bias-variance identity:** for any target θ and estimator $\hat{\theta}$,
 $\mathbb{E}[(\hat{\theta} - \theta)^2] = (\mathbb{E}[\hat{\theta}] - \theta)^2 + \text{Var}(\hat{\theta})$.

7.3 Where Each Idea Reappears

Forward pointers

- **Lecture 11 (non-parametric methods)**. Basis expansions add polynomial and other transformed predictor columns to build flexible curves; basis size controls the flexibility.
- **Lecture 12 (logistic regression)**. The same penalised loss-minimisation template applied to a binary response. Ridge-logistic and LASSO-logistic are routine.
- **Lecture 13 (cross-validation)**. The procedure that picks λ from data — the missing piece that turns ridge and LASSO into a complete recipe.

One last reminder. OLS is not wrong — it is the right tool when $n \gg p + 1$. Ridge and LASSO are the right tools when it is not.